

Unnamed_Unit

Unit 2 - Understanding Angles with IMU

PDF Documents

PDF Document 1: 1739430053546-Chapter 4 - unit 2 - Understanding Angles with IMU.pdf

This PDF document is included in the unit materials.



UNIT 2 UNDERSTANDING ANGLES WITH IMU

WHAT IS IMU?



IMU stands for Inertial Measurement Unit. It's a special tool inside devices like smartphones that helps them understand how they are moving. Think of it as a tiny, smart sensor that can feel which way you tilt, turn, or move your phone.

HOW DOES IMU WORK?

Imagine you are playing a car racing game on your phone. When you turn the phone left, the car on the screen turns left. When you tilt the phone forward, the car goes faster. The IMU is what makes this happen! It senses the angle you hold your phone and sends this information to the game.



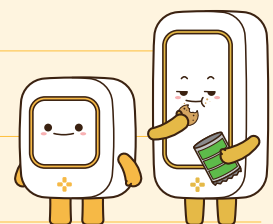
● Learning About Angles with IMU

The IMU helps the phone know about three types of movements:

- 1 ⇔ **Tilt Left or Right:** This is like turning a steering wheel left or right.
- 2 ⇕ **Tilt Forward or Backward:** This is like speeding up or slowing down.

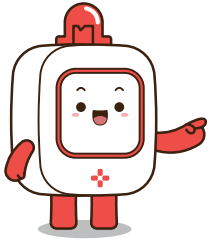
THINK ABOUT ELSE

What are other things that can move forwards, backwards, and side to side?



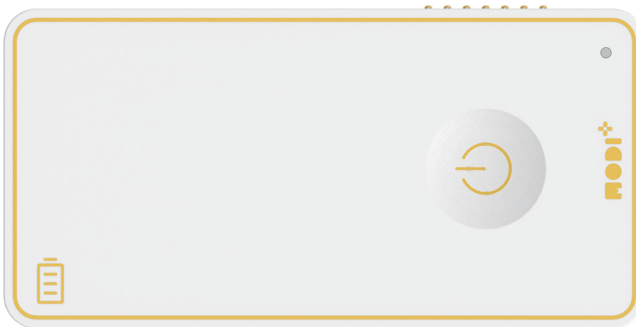


HOW DOES IMU WORK?



In this activity, we will use a MODI kit to learn about angles and how movements are detected. We will connect a battery, an IMU (Inertial Measurement Unit), and a display. The display will show how the IMU senses left and right movements.

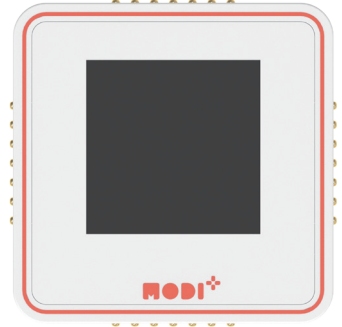
MATERIALS NEEDED:



Battery

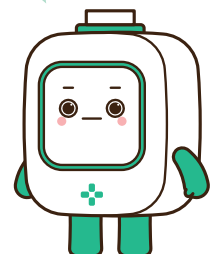


IMU



Display

Try connecting these three modules using the magnetic parts!

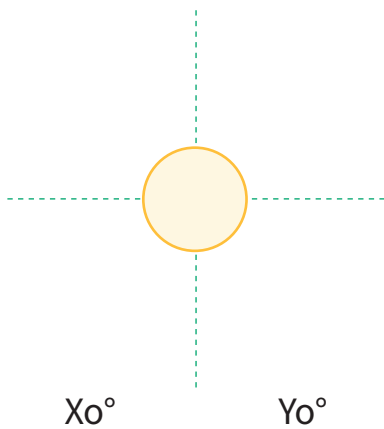


Let's figure it out

Draw what appears on the display as you move the IMU.

When the IMU is on desk

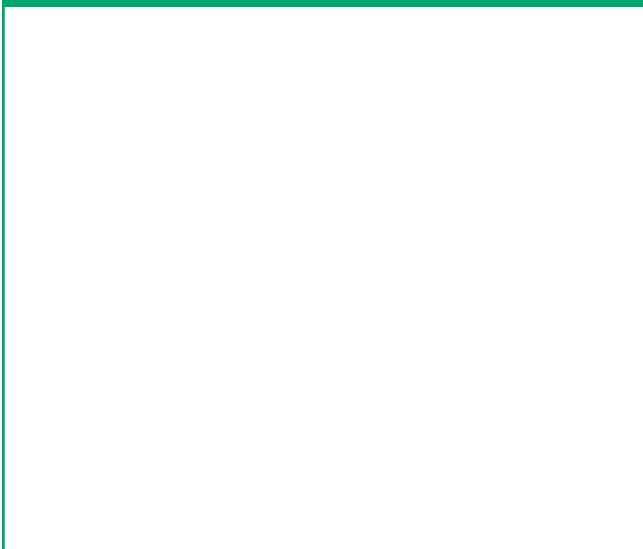
For example..



When the IMU is standing upright



When tilted to the left



When tilted to the right

